PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf)** and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Comparison of conventional, Doppler, and contrast-enhanced ultrasonography in differential diagnosis of ovarian masses: a systematic review and meta-analysis
AUTHORS	Xun, Lizhang; Zhai, Lamei; Xu, Hui

VERSION 1 – REVIEW

REVIEWER	Kannan Sridharan Seth GS Medical College and KEM Hospital, Department of Clinical Pharmacology
REVIEW RETURNED	18-May-2021

GENERAL COMMENTS	Please GRADE the pooled estimates of the outcomes.

REVIEWER	M Pal UPES, School of Health Sciences
REVIEW RETURNED	28-Jul-2021

GENERAL COMMENTS	Comparison of Conventional, Doppler, and Contrast-Enhanced Ultrasonography in Differential Diagnosis of Ovarian Masses: A Systematic Review and Meta-analysis (Manuscript ID: bmjopen-2021-052830) Comments to the author: The manuscript titled “Comparison of Conventional, Doppler, and Contrast-Enhanced Ultrasonography in Differential Diagnosis of Ovarian Masses: A Systematic Review and Meta-analysis” submitted by Xun et al., investigated on the comparison study on the value of conventional, Doppler and CEUS methods for differential diagnosis of benign and malignant ovarian cancer masses. The authors have also explained the overall performance of CEUS, a reliable and convenient method, is superior to other methods. Discrimination of ovarian lesions is primary importance in gynaecological practice. Herein, the authors have done well to compare different diagnostic methods based on meta-analysis. However, the manuscript needs to do minor modification before acceptance. My general observations are as follows. Minor comments: 1. Ovarian tumours are typically categorized as epithelial, germ cell, sex cord–stromal, or metastatic, whereas epithelial tumours are the most common type (Nat Rev Cancer. 2017 Jan;17(1):65-74.; Int. J. Biol. Macromol. 2018, 118(Pt A):1326-1345; Radiographics. 2000 Sep-Oct; 20(5):1445-70; Experimental Cell Research, 2017, 362(1):1-10). So the authors need to mention clearly about the types and subtypes (serous, mucinous, endometrioid, clear cell and Brenner tumours) they have focused on this article in details.

	2. Authors may also address in discussion: How does combining Doppler imaging with morphologic assessment improve diagnostic accuracy of ovarian cancer? 3. Both conventional and Doppler US are comparatively less accurate than magnetic resonance (MR). So authors also need to mention, at least in few instances with examples, the sensitivity and specificity of CEUS how close to the most accurate method MR imaging approaches in diagnosing ovarian malignancy.

REVIEWER	Katja Glutig University Hospital Jena Clinic and Polyclinic of ORL, Institute for Diagnostic and Interventional Radiology
REVIEW RETURNED	21-Oct-2021

GENERAL COMMENTS	This is a well-written article that identifies an important gap. The subject is very important. It's currently something of a "hot topic", and is one to which the author has made significant contributions. 60 Annually, over 200,000 women in the USA undergo surgical interventions for pelvic masses; moreover, approximately 22,240 ovarian cancer cases were diagnosed in 2018 This reference does not seem very up-to-date. Is there a current guideline recommendation on this? 70 Conventional US can visualize the capsule and tumor shapes, which could allow differential diagnosis of benign or malignant tumors [9]. This reference seems inappropriate for the statement made.Computed Ultrasound Risk Evaluation (CURE) system has been designed with the clinical goals of whole breast, operator-independent imaging, and differentiation of breast masses. This is a further development of fundamental ultrasound, especially for breast diagnostics. I would recommend another Reference. 74 Furthermore, contrast- enhanced US (CEUS) could improve imaging quality [12]. This reference does not seem up to date to me either. There exist EFSUMB Guidelines and Recommendations for the Clinical Practice of Contrast-Enhanced Ultrasound (CEUS) in Non-Hepatic Applications. I would prefer the following reference at this point: Sidhu PS, Cantisani V, Dietrich CF, Gilja OH, Saftoiu A, Bartels E, Bertolotto M, Calliada F, Clevert DA, Cosgrove D, Deganello A, D'Onofrio M, Drudi FM, Freeman S, Harvey C, Jenssen C, Jung EM, Klauser AS, Lassau N, Meloni MF, Leen E, Nicolau C, Nolsoe C, Piscaglia F, Prada F, Prosch H, Radzina M, Savelli L, Weskott HP, Wijkstra H. The EFSUMB guidelines and recommendations for the clinical practice of contrast-enhanced ultrasound (CEUS) in non-hepatic applications: Update 2017 (long version). Ultraschall Med 2018b;39:e2–e44. 91 The following search terms were used as text words or Medical Subject Heading terms: "ovarian neoplasms" AND ("ultrasonography" OR "Doppler ultrasonography" OR "contrast-enhanced ultrasonography") AND "diagnosis." I am surprised that there are not many references in the last included years 2016 to 2000. The 2016, 2017 and 2018 data are from Thailand, Bangladesh and Iraq. Is there really no more recent data from Europe, US, or China? 98 (2) Participants: patients with ovarian masses I would very clearly like the age of the patients to be stated. What age range was set as the inclusion criterion? Are data from paediatric cases included? If not, please briefly mention where I assume this to be the case.

	147 (Supplemental 2) I can see these results in Table 2. 158 (Supplemental 3) I can see these results in Table 2. 159 Conventional US and Doppler US had a higher and lower NLR than CEUS (ratio: 3.33; 95%CI: 1.04-1 0.66; P = 0.042) and (ratio: 0.53; 95%CI: 0.28-1.00; P = 0.050), respectively (Table 2). (ratio: 0.53; 95%CI: 0.28-1.00; P = 0.050) I see these values in Table 2 for PLR in comparison Doppler US versus CEUS. Please verify. 166 (Supplemental 4) I can see these results in Table 2. 176 (Figure 2) I can see these results in Table 2. 200 Discussion: To me, this 2007 reference by Liu J seems outdated. In my opinion, a more up-to-date reference is more helpful here or that could be evaluated more precisely here. 218 Moreover, the blood flow velocity in Doppler US could differentiate between benign and malignant pelvic masses [32]. This statement from the reference by Kurtz AB, Tsimikas JV, Tempany CM, et al. Diagnosis and staging of ovarian cancer: comparative values of Doppler and conventional US, CT, and MR imaging correlated with surgery and histopathologic analysisVreport of the Radiology Diagnostic Oncology Group. Radiology 1999;212:19-27 I find extremely difficult. The reference is outdated in my opinion. Data from 1999 do not reflect current ultrasound standards. Differentiation at this point would be helpful. 222 This discussion could be expanded to explain the advantages of contrast-enhanced ultrasound more detail. I would find further current references helpful at this point. The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

VERSION 1 – AUTHOR RESPONSE

Response to editor

Question 1: Might the authors considering updating their search, in case additional relevant studies have been published since the search was performed?

Response: Thanks for this suggestion. The literature search has already been updated in October 2021, while no new eligible study met the inclusion criteria. We found mostly studies illustrated the diagnostic model on the basis of ultrasound features and the combination of gynecologic imaging reporting and data system. The PRISMA Flowchart have already updated in Figure 1.

Question 2: Please revise the formatting of your abstract so that it includes the following sections: Objectives >> Design >> Data sources >> Eligibility criteria >> Data extraction and synthesis >> Results >> Conclusions. Please see the following published abstract as an example: http://bmjopen.bmj.com/content/8/3/e019438.

Response: Thanks for this suggestion, and the formatting of abstract have already changed in the revised manuscript.

Question 3: Please ensure that the abstract contains an adequate description of the search strategy and selection criteria. The “Data sources” section should list all the databases searched and the date range covered by the search. The “Eligibility criteria” section should include language restrictions as well as key inclusion and exclusion criteria.

Response: Thanks for this suggestion, and these sections have already added in Abstracts section.

Question 4: Please update the Results section of the abstract to include measures of significance (eg, 95% CIs) for all reported findings.

Response: Thanks for this suggestion, and measures of significance (eg, 95% CIs) for all reported findings have already added in the revised manuscript.

Question 5: Please revise the ‘Strengths and limitations’ section of your manuscript (after the abstract). This section should contain up to five short bullet points, no longer than one sentence each, that relate specifically to the methods. The overall study design and results of the study should not be summarised here.

Response: Thanks for this suggestion, and the ‘Strengths and limitations’ section have already changed in the revised manuscript.

Question 6: In your main text Methods, please indicate which two authors (by initials) did the search and study selection, which two authors did the data collection and quality assessment, and which one author resolved inconsistencies in the latter.

Response: Thanks for this suggestion, and the initials of authors have already added in the revised manuscript.

Question 7: Please replace the PRISMA 2009 checklist with a completed copy of the updated PRISMA 2020 checklist, updating the manuscript as needed to ensure all reporting requirements are met.

Response: Thanks for this suggestion, and the PRISMA checklist have already replaced in the revised manuscript.

Response to Reviewer #1

General comments: Please GRADE the pooled estimates of the outcomes.

Response: Thanks for this suggestion. The pooled estimates of outcomes with GRADE mainly focused on meta-analysis of randomized controlled trials. Our study on the basis of diagnostic studies, and the quality of included studies are shown in Supplemental 1.

Response to Reviewer #2

General comments: The manuscript titled “Comparison of Conventional, Doppler, and Contrast-Enhanced Ultrasonography in Differential Diagnosis of Ovarian Masses: A Systematic Review and Meta-analysis” submitted by Xun et al., investigated on the comparison study on the value of conventional, Doppler and CEUS methods for differential diagnosis of benign and malignant ovarian cancer masses. The authors have also explained the overall performance of CEUS, a reliable and convenient method, is superior to other methods.

Discrimination of ovarian lesions is primary importance in gynaecological practice. Herein, the authors have done well to compare different diagnostic methods based on meta-analysis. However, the manuscript needs to do minor modification before acceptance. My general observations are as follows.

Response: As behalf of all co-authors, I would like to appreciate this referee due to thoughtful comments proposed by the peer review. After we revised the manuscript, those significant issues could be changed.

Question 1: Ovarian tumours are typically categorized as epithelial, germ cell, sex cord–stromal, or metastatic, whereas epithelial tumours are the most common type (Nat Rev Cancer. 2017 Jan;17(1):65-74.; Int. J. Biol. Macromol. 2018, 118(Pt A):1326-1345; Radiographics. 2000 Sep-Oct; 20(5):1445-70; Experimental Cell Research, 2017, 362(1):1-10). So the authors need to mention clearly about the types and subtypes (serous, mucinous, endometrioid, clear cell and Brenner tumours) they have focused on this article in details.

Response: Thanks for this suggestion, and the type of ovarian cancer have already added in Table 1. Moreover, we have already addressed this comment in Limitation section.

Question 2: Authors may also address in discussion: How does combining Doppler imaging with morphologic assessment improve diagnostic accuracy of ovarian cancer?

Response: Thanks for this suggestion, and the following sentences have already added in the revised manuscript: “the use of Doppler US to assess the gray-scale ultrasound morphology in an adnexal mass with high accurate for predicting its nature”.

Question 3: Both conventional and Doppler US are comparatively less accurate than magnetic resonance (MR). So authors also need to mention, at least in few instances with examples, the

sensitivity and specificity of CEUS how close to the most accurate method MR imaging approaches in diagnosing ovarian malignancy.

Response: Thanks for this suggestion, and the diagnostic value of CEUS as compared with MR in prior study have already illustrated in Discussion section.

Response to Reviewer #3

General comments: This is a well-written article that identifies an important gap. The subject is very important. It's currently something of a 'hot topic', and is one to which the author has made significant contributions.

Response: As behalf of all co-authors, I would like to appreciate this referee due to thoughtful comments proposed by the peer review. After we revised the manuscript, those significant issues could be changed.

Question 1: 60 Annually, over 200,000 women in the USA undergo surgical interventions for pelvic masses; moreover, approximately 22,240 ovarian cancer cases were diagnosed in 2018

This reference does not seem very up-to-date. Is there a current guideline recommendation on this?

Response: Thanks for this suggestion, and the reference have already updated in the revised manuscript.

Question 2: 70 Conventional US can visualize the capsule and tumor shapes, which could allow differential diagnosis of benign or malignant tumors [9].

This reference seems inappropriate for the statement made.Computed Ultrasound Risk Evaluation (CURE) system has been designed with the clinical goals of whole breast, operator- independent imaging, and differentiation of breast masses. This is a further development of fundamental ultrasound, especially for breast diagnostics. I would recommend another Reference.

Response: Thanks for this suggestion, and this reference have already changed in the revised manuscript.

Question 3: 74 Furthermore, contrast- enhanced US (CEUS) could improve imaging quality [12].

This reference does not seem up to date to me either. There exist EFSUMB Guidelines and Recommendations for the Clinical Practice of Contrast-Enhanced Ultrasound (CEUS) in Non-Hepatic Applications. I would prefer the following reference at this point:

Sidhu PS, Cantisani V, Dietrich CF, Gilja OH, Saftoiu A, Bartels E, Bertolotto M, Calliada F, Clevert DA, Cosgrove D, Deganello A, D'Onofrio M, Drudi FM, Freeman S, Harvey C, Jenssen C, Jung EM, Klauser AS, Lassau N, Meloni MF, Leen E, Nicolau C, Nolsoe C, Piscaglia F, Prada F, Prosch H,

Radzina M, Savelli L, Weskott HP, Wijkstra H. The EFSUMB guidelines and recommendations for the clinical practice of contrast-enhanced ultrasound (CEUS) in non-hepatic applications: Update 2017 (long version). Ultraschall Med 2018b;39:e2–e44.

Response: Thanks for this suggestion, and this reference have already changed in the revised manuscript.

Question 4: 91 The following search terms were used as text words or Medical Subject Heading terms: “ovarian neoplasms” AND (“ultrasonography” OR “Doppler ultrasonography” OR “contrast-enhanced ultrasonography”) AND “diagnosis.”

I am surprised that there are not many references in the last included years 2016 to 2000. The 2016, 2017 and 2018 data are from Thailand, Bangladesh and Iraq. Is there really no more recent data from Europe, US, or China?

Response: Thanks for this suggestion, and the literature search have already updated in the revised manuscript.

Question 5: 98 (2) Participants: patients with ovarian masses

I would very clearly like the age of the patients to be stated. What age range was set as the inclusion criterion? Are data from paediatric cases included? If not, please briefly mention where I assume this to be the case.

Response: Thanks for this suggestion, and this sentence have already changed into: “adult women experience ovarian masses”.

Question 6: 147 (Supplemental 2)

I can see these results in Table 2.

Response: Thanks for this suggestion. Although the results in Table 2 were consistent with Supplemental 2, while Figures in Supplemental 2 could display the result in individual study.

Question 7: 158 (Supplemental 3)

I can see these results in Table 2.

Response: Thanks for this suggestion. Although the results in Table 2 were consistent with Supplemental 3, while Figures in Supplemental 3 could display the result in individual study.

Question 8: 159 Conventional US and Doppler US had a higher and lower NLR than CEUS (ratio: 3.33; 95%CI: 1.04-1 0.66; P = 0.042) and (ratio: 0.53; 95%CI: 0.28-1.00; P = 0.050), respectively (Table 2).

(ratio: 0.53; 95%CI: 0.28-1.00; P = 0.050) I see these values in Table 2 for PLR in comparison Doppler US versus CEUS. Please verify.

Response: Thanks for this suggestion, and this sentence have already changed into: “Conventional US versus CEUS showed higher NLR (ratio: 3.33; 95%CI: 1.04-10.66; P = 0.042), while Doppler US versus CEUS showed lower PLR (ratio: 0.53; 95%CI: 0.28-1.00; P = 0.050) (Table 2).”

Question 9: 166 (Supplemental 4)

I can see these results in Table 2.

Response: Thanks for this suggestion. Although the results in Table 2 were consistent with Supplemental 4, while Figures in Supplemental 4 could display the result in individual study.

Question 10: 176 (Figure 2)

I can see these results in Table 2.

Response: Thanks for this suggestion. Although the results in Table 2 were consistent with Figure 2, while Figure 2 could display the result in individual study.

Question 11: 200 Discussion: To me, this 2007 reference by Liu J seems outdated. In my opinion, a more up-to-date reference is more helpful here or that could be evaluated more precisely here.

Response: Thanks for this suggestion, and this sentence have already removed in the revised manuscript.

Question 12: 218 Moreover, the blood flow velocity in Doppler US could differentiate between benign and malignant pelvic masses [32].

This statement from the reference by Kurtz AB, Tsimikas JV, Tempany CM, et al. Diagnosis and staging of ovarian cancer: comparative values of Doppler and conventional US, CT, and MR imaging correlated with surgery and histopathologic analysisVreport of the Radiology Diagnostic Oncology Group. Radiology 1999;212:19-27 I find extremely difficult. The reference is outdated in my opinion. Data from 1999 do not reflect current ultrasound standards. Differentiation at this point would be helpful.

Response: Thanks for this suggestion, and the reference have already updated in the revised manuscript.

Question 13: 222 This discussion could be expanded to explain the advantages of contrast-enhanced ultrasound more detail. I would find further current references helpful at this point.

Response: Thanks for this suggestion, and several sentences have already added in Discussion section to address your comment.

VERSION 2 – REVIEW

REVIEWER	Katja Glutig University Hospital Jena Clinic and Polyclinic of ORL, Institute for

	Diagnostic and Interventional Radiology
REVIEW RETURNED	18-Nov-2021

GENERAL COMMENTS	Many thanks to the authors. My comments have been taken into account in the revised manuscript. I consider the work comprehensive and am in favor of acceptance.